ELECTRONICS AND COMMUNICATION ENGINEERING

BEC -501	INTEGRATED CIRCUITS	3L:1T:0P	4 Credits

Unit	Topics	Lectures
I	**The 741 IC Op-Amp**: Introduction to integrated circuits, types of integrated circuits, Device parameters (properties of Opamp), slew rate, relationship between frequency and slew rate, gain, frequency response of 741, a simplified equivalent model of opamp, inverting amplifier, non-inverting amplifier. Basic Applications of Opamp.	8
II	**Linear Applications of IC Op-Amps:** Op-Amp based V-I and I-V converters, instrumentation amplifier, generalized impedance converter, simulation of inductors. **Active Analog filters:** First order active filters, Low pass Filter design, practical low pass filter design, High pass filter design, second order filters, implementation topologies, Designing of second order lowpass and high pass Butterworth filter, Introduction to band pass and band stop filter, all pass active filters, Sallen Key second order filter, KHN Filters. Introduction to design of higher order filters.	8
III	**Non-Linear Applications of IC Op-Amps:** Basic Log–Anti Log amplifiers using diode and BJT, temperature compensated Log-Anti Log amplifiers using diode, Analog Multiplier circuit design using log amplifier, Analog divider circuit design using log amplifier, peak detectors, sample and hold circuits. Op-amp as a comparator and zero crossing detector, Schmitt triggers, Hysteresis voltage, Hysteresis curve, Astable multivibrator & monostable multivibrator, Generation of square waveform, Generation of triangular waveform. Frequency of the generated triangular waveform.	4 8
IV	**Digital Integrated Circuit Design:** An overview of digital integrated circuits, Basic logic generation in ICs, CMOS logic gate circuit's basic structure, CMOS realization of inverters, AND, OR, NAND and NOR gates. **Latches and Flip flops**: the latch, CMOS implementation of SR flip-flops, a simpler CMOS implementation of the clocked SR flip-flop, CMOS implementation of J-K flip- flops, D flip- flop circuits.	6
V	**Integrated Circuit Timer:** Introduction to 555 Timer IC, Timer IC 555 pin and functional block diagram, Internal Architecture, Operating Modes of 555 Timer: Monostable and Astable multivibrator using the 555 IC, Bistable multivibrator using IC 555. **Applications:** Pulse Width Modulation (PWM) with 555, Ramp/triangle wave generator,	6

Text Book:

1. Microelectronic Circuits, Sedra and Smith, 7th Edition, Oxford, 2017.
2. Behzad Razavi: Design of Analog CMOS Integrated Circuits, TMH

Reference Books:

1. Gayakwad: Op-Amps and Linear Integrated Circuits, 4th Edition Prentice Hall of India, 2002.
2. Franco, Analog Circuit Design: Discrete & Integrated, TMH, 1st Edition.
3. Salivahnan, Electronics Devices and Circuits, TMH, 3rd Edition, 2015
4. Millman and Halkias: Integrated Electronics, TMH, 2nd Edition, 2010

Course Outcomes: At the end of this course students will demonstrate the ability to:

1. Explain complete internal analysis of Op-Amp 741-IC.
2. Examine and design Op-Amp based circuits and basic components of ICs such as various types of filters.
3. Implement the concept of Op-Amp to design Op-Amp based non-linear applications.
4. Analyse and design basic digital IC circuits using CMOS technology.
5. Describe the functioning of application specific ICs such as 555 timer, PWM and ramp generators.